EXECUTIVE BRIEFING

Cut your agentic-coding token bill

Coding agents are becoming a top-line AI cost. The lever is choosing the right **model and harness** from evidence -- not defaulting to the most expensive option.

THE PROBLEM

Organizations are heading toward trillions of tokens a month -- and coding agents drive a large, growing share of it.

Two choices decide most of that bill -- **which agent (harness)** drives the work and **which model** it uses -- yet they are usually made on gut feel or on public leaderboards that may not reflect your real work.

WHY GENERIC NUMBERS MISLEAD

Real coding work doesn't look like a benchmark prompt

50-300

LLM turns per task -- long-horizon, not one-shot

150:1+

input-to-output ratio -- read-heavy, up to 660:1

3×

cost swing on the same model, just from the harness

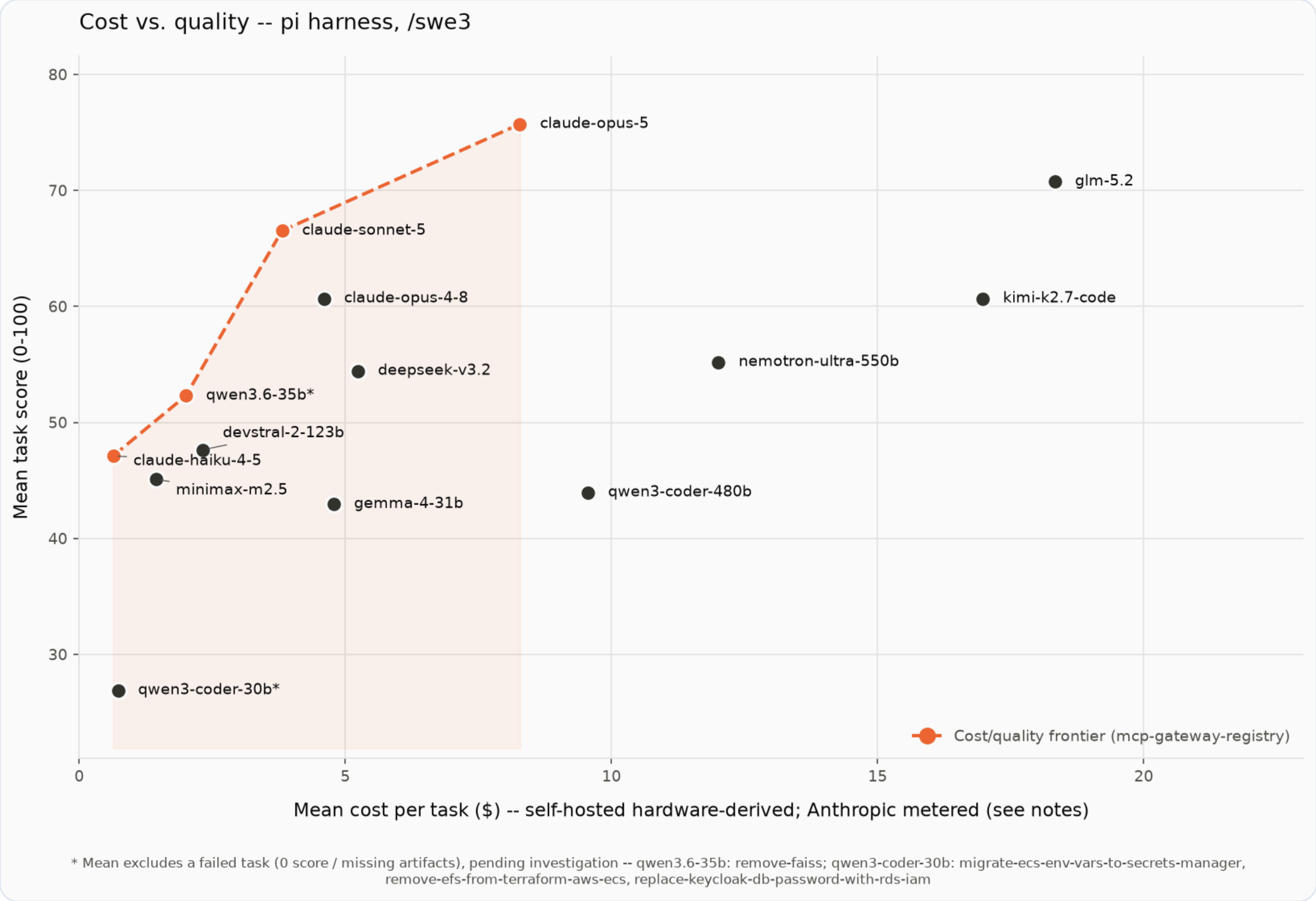
Cost depends on **how** a model drives the task. You need numbers measured on **work that looks like yours**.

THE APPROACH

Measure quality and cost on real coding tasks -- across agents and hosting options -- then let the cost/quality frontier pick the model.

The frontier is the set of models where nothing else is both better and cheaper.
It turns model selection into a defensible, budget-aware decision.

Your buying guide: the cost/quality frontier



14 models on real tasks. Higher is better; left is cheaper. The line is the frontier -- the only models worth considering. Everything below it is dominated (something is both better and cheaper).

WHAT TO BUY FOR WHAT

Match the model to the job

Business-critical work -- top-quality frontier model. Pay the most, get the most; use it where a wrong result is expensive.

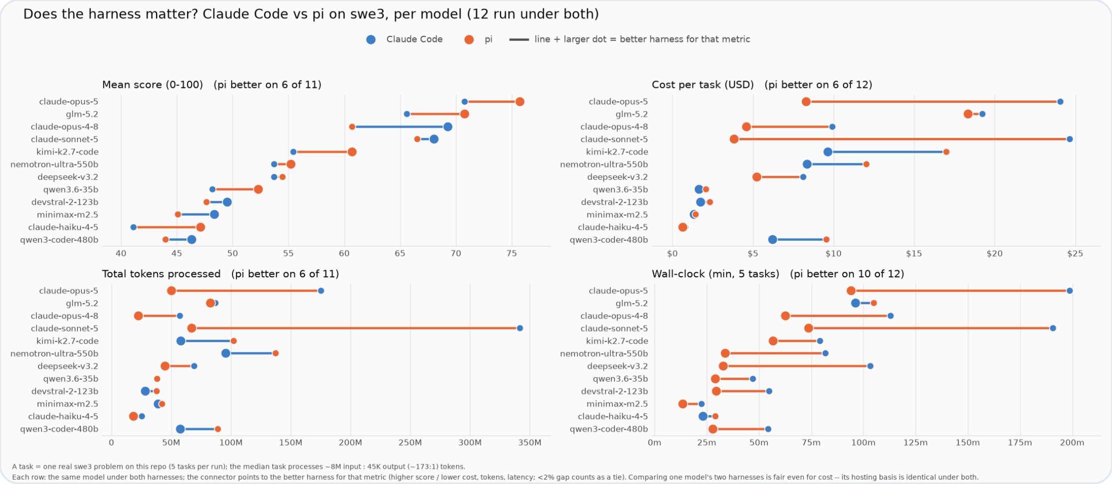
Everyday engineering -- a workhorse model at a fraction of the cost delivers most of the quality for the bulk of tickets.

Routine / high-volume edits -- the most cost-effective model handles boilerplate and small fixes you review anyway.

The endgame: a harness that **routes each task to the right tier automatically** -- frontier quality at a fraction of the spend, with no one managing model choice.

THE OVERLOOKED LEVER

The agent matters as much as the model



Same model, different agent = up to 3× the cost and double the wall-clock, for little or no quality gain.

A leaner agent won faster turnaround on 10 of 12 models -- and tied or beat on cost.

Buy optionality: the freedom to pair each model with the cheaper agent is itself a cost lever.

BEDROCK OR SELF-HOSTED

Works whether you buy tokens on Amazon Bedrock or self-host open-weight models on your own GPUs.

Same tasks, same scoring -- so the trade-off is measured on one footing. At high volume, self-hosting a right-sized open-weight model on committed GPU capacity is a real lever against per-token pricing.

THE TAKEAWAY

Turn coding-agent spend into a managed decision

Stop defaulting to the most expensive model-and-agent for every task.

Choose from the frontier -- measured on your repos, your tasks, your hosting.

Route by tier -- and capture a large cost reduction with little quality give-up.

EXPLORE THE EVIDENCE

See the full benchmark

Open source. Reproduce the frontier on your own code and models.

github.com/aarora79/agentic-coding-harness-benchmarks

Scan for the repository